

Lecture 10: From Geometric Satake to Geometric Langlands

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Where We Are

Lectures 1–9: Proved the geometric Satake equivalence

$$\mathrm{Perv}_{L^+G}(\mathrm{Gr}_G) \simeq \mathrm{Rep}(\hat{G})$$

as a symmetric monoidal equivalence.

Recall: $LG = G(k((t)))$ (loop group), $L^+G = G(k[[t]])$ (positive loops), $\mathrm{Gr}_G = LG/L^+G$ (affine Grassmannian). $\mathrm{Perv}_{L^+G}(\mathrm{Gr}_G)$ = category of L^+G -equivariant perverse sheaves on Gr_G . $\hat{G}$ = Langlands dual group (root datum dual to G).

Today: What does this equivalence **mean**?

- How does it fit into the Langlands program?
- What does it generalize to?
- Where does it lead?

1. **Classical Satake as LLC** — the Satake isomorphism is the unramified case of local Langlands
2. **Geometric Satake and generalizations** — from Gr to Fl , from Tannakian to categorical
3. **Global Geometric Langlands** — eigensheaves on Bun_G
4. **Big picture** — the level hierarchy and open problems

Conventions

- F : non-archimedean local field ($\mathbb{Q}_p$ or $\mathbb{F}_q((t))$), $\mathcal{O}_F$, $\mathbb{F}_q$, uniformizer ϖ
- W_F : Weil group; $I_F \subset W_F$: inertia; $\text{Frob} \in W_F/I_F$: geometric Frobenius
- ${}^L G = \hat{G} \rtimes \text{Gal}(\bar{F}/F)$: the L -group
- $L^+ G = G(\mathcal{O}_F)$, $LG = G(F)$, $\text{Gr}_G = LG/L^+ G$
- X : smooth projective curve over $\mathbb{F}_q$ (or $\mathbb{C}$); $F_X = \mathbb{F}_q(X)$

Classical Satake as Local Langlands

The L -group

The L -group of G sits in a short exact sequence

$$1 \longrightarrow \hat{G} \longrightarrow {}^L G \longrightarrow \mathrm{Gal}(\bar{F}/F) \longrightarrow 1$$

$\hat{G}$ = Langlands dual group (root datum dual to G).

Semi-direct product: ${}^L G = \hat{G} \rtimes \mathrm{Gal}(\bar{F}/F)$, where $\sigma \in \mathrm{Gal}$ acts on $\hat{G}$ via the automorphism induced by σ 's action on $X^*(T) = X_*(\hat{T})$.

When G is **split** (as in this seminar): action is trivial, so
 ${}^L G = \hat{G} \times \mathrm{Gal}(\bar{F}/F)$.

Definition (L-parameter)

A continuous homomorphism

$$\varphi : W_F \times \mathrm{SL}_2(\mathbb{C}) \longrightarrow {}^L G = \hat{G} \rtimes \mathrm{Gal}(\bar{F}/F)$$

satisfying:

1. Projection to $\mathrm{Gal}(\bar{F}/F)$ agrees with $W_F \rightarrow \mathrm{Gal}(\bar{F}/F)$
2. $\varphi|_{W_F}$ is smooth with semisimple image
3. $\varphi|_{\mathrm{SL}_2}$ is algebraic

Two L-parameters are equivalent if $\hat{G}$ -conjugate.

Three Pieces of Data in φ

The domain $W_F \times \mathrm{SL}_2(\mathbb{C})$ has three parts:

1. **Frobenius**: $\varphi(\mathrm{Frob}) \in {}^L G$ — semisimple element = **Satake parameter**
(when φ is unramified, this is the only data)
2. **Inertia**: $\varphi|_{I_F} : I_F \rightarrow \hat{G}$ — records **ramification**
3. **Monodromy**: $\varphi|_{\mathrm{SL}_2} : \mathrm{SL}_2(\mathbb{C}) \rightarrow \hat{G}$ — records **monodromy operator**
 $N := \log \varphi\left(\begin{smallmatrix} 1 & \\ 0 & 1 \end{smallmatrix}\right) \in \hat{\mathfrak{g}}$ is nilpotent. $N = 0$ means “no monodromy.”

Unramified: $\varphi|_{I_F} = 1$ and $N = 0$, so φ is determined by $\varphi(\mathrm{Frob})$ alone.

Centralizer, Component Group, and LLC

Definition (Centralizer and component group)

For an L-parameter φ :

$$C_\varphi = \{g \in \hat{G} : g\varphi(w)g^{-1} = \varphi(w) \forall w \in W_F \times \mathrm{SL}_2\}$$
$$S_\varphi = \pi_0(C_\varphi/Z(\hat{G})^{\mathrm{Gal}})$$

Notation: $\mathrm{Irr}(H)$ = set of isomorphism classes of irreducible (smooth, for p -adic groups; algebraic, for finite groups) representations of H .

Conjecture (Local Langlands Correspondence)

There exists a partition

$$\mathrm{Irr}(G(F)) = \bigsqcup_{\varphi} \Pi_{\varphi}$$

into finite **L-packets**, with injections $\iota : \Pi_{\varphi} \hookrightarrow \mathrm{Irr}(S_{\varphi})$.

For $G = \mathrm{GL}_n$: $S_{\varphi} = 1$ always $\Rightarrow$ L-packets are singletons $\Rightarrow$ LLC is a bijection

Example: GL_2 Centralizer Computation

$G = \mathrm{GL}_2$, $\varphi = \chi_1 \oplus \chi_2$ with $\chi_1 \neq \chi_2$:

$$C_\varphi = \{g \in \mathrm{GL}_2(\mathbb{C}) : g \text{ commutes with } \mathrm{diag}(\chi_1(w), \chi_2(w)) \forall w\} = \{\mathrm{diag}(a, b)\}$$

$$S_\varphi = \pi_0((\mathbb{C}^\times)^2 / \mathbb{C}^\times) = 1$$

For $\chi_1 = \chi_2$: $C_\varphi = \mathrm{GL}_2(\mathbb{C})$, still $S_\varphi = 1$.

In fact, $S_\varphi = 1$ for **all** L-parameters of GL_n .

Theorem (Harris–Taylor, Henniart; Laumon–Rapoport–Stuhler). *LLC holds for GL_n over any non-archimedean local field. Each L-packet is a singleton.*

The Classical Satake Isomorphism

$K = G(\mathcal{O}_F)$ hyperspecial. Spherical Hecke algebra:

$$\mathcal{H}(G, K) = C_c^\infty(K \backslash G(F) / K).$$

Theorem (Satake)

$$\mathcal{S} : \mathcal{H}(G, K) \xrightarrow{\sim} \mathbb{C}[X_*(T)]^W \cong R(\hat{G})$$

$$\text{Explicitly: } (Sf)(t) = \delta_B(t)^{1/2} \int_U f(tu) du,$$

$$\delta_B(t) = |\det(\text{Ad}(t)|_{\text{Lie}(U)})|_F.$$

Example ($G = \text{GL}_2$, $F = \mathbb{Q}_p$):

$$\mathcal{S} : \mathcal{H}(\text{GL}_2, K) \xrightarrow{\sim} \mathbb{C}[z_1^{\pm 1}, z_2^{\pm 1}]^{S_2}$$

- $\mathcal{S}(1_{K \text{diag}(p,1)K}) = p^{1/2}(z_1 + z_2)$ — character of std
- $\mathcal{S}(1_{K \text{diag}(p,p)K}) = z_1 z_2$ — character of det

Satake = Unramified LLC

Proposition. An unramified L-parameter ($\varphi|_{I_F} = 1$, $\varphi|_{\mathrm{SL}_2} = 1$) is determined by $\varphi(\mathrm{Frob}) \in \hat{G}^{\mathrm{ss}}/\mathrm{conj}$.

Why: $W_F/I_F \cong \mathbb{Z}$, generated by Frob . Since $\varphi|_{I_F} = 1$, φ factors through $\mathbb{Z}$, determined by $\varphi(\mathrm{Frob})$.

Corollary (Borel)

$$\{\text{irred. spherical reps of } G(F)\} \cong \{\text{unramified L-parameters}\} / \hat{G}\text{-conj}$$

The bijection: $\pi \mapsto$ eigenvalue of $\mathcal{H}(G, K)$ on π^K . Via $\mathcal{H}(G, K) \cong R(\hat{G})$, the eigenvalue is a maximal ideal = semisimple conjugacy class = $\varphi_\pi(\mathrm{Frob})$.

Why Satake = Unramified LLC: The Square

$$\begin{array}{ccc}
 \{\text{irred. spherical } \pi\} & \xrightarrow{\sim} & \{\text{unram. } \varphi\} / \hat{G} \\
 \pi \mapsto \pi^K \downarrow & & \downarrow \varphi \mapsto \varphi(\text{Frob}) \\
 \{\text{irred. } \mathcal{H}(G, K)\text{-mod}\} & \xrightarrow[\mathcal{S}^*]{\sim} & \hat{T}/W
 \end{array}$$

Left: $\pi \mapsto \pi^K$ is an equivalence (spherical reps $\leftrightarrow \mathcal{H}(G, K)$ -modules).

Bottom: Satake $\mathcal{H}(G, K) \xrightarrow{\sim} \mathbb{C}[X_*(T)]^W \cong \mathbb{C}[\hat{T}/W]$. An irred. module = maximal ideal of $\mathbb{C}[\hat{T}/W]$ = point of $\hat{T}/W$.

Right: unramified $\varphi(\text{Frob}) \in \hat{G}^{\text{ss}}/\text{conj} \cong \hat{T}/W$ (Chevalley restriction).

Top: LLC for unramified reps = composite of left, bottom, right⁻¹.

Beyond Satake: Ramified Principal Series

$\chi_1, \chi_2 : \mathbb{Q}_p^\times \rightarrow \mathbb{C}^\times$ with some χ_i **ramified** ($\chi_i|_{\mathbb{Z}_p^\times} \neq 1$).

$$\pi(\chi_1, \chi_2) = \text{Ind}_B^{\text{GL}_2}(\chi_1 \otimes \chi_2), \quad \pi(\chi_1, \chi_2)^K = 0$$

L-parameter: $\varphi = \chi_1 \oplus \chi_2 : W_{\mathbb{Q}_p} \rightarrow \text{GL}_2(\mathbb{C})$

Since χ_i ramified: $\varphi|_{I_{\mathbb{Q}_p}} \neq 1$, but $\varphi|_{\text{SL}_2} = 1$.

Invisible to $\mathcal{H}(G, K)$, but visible to $\mathcal{H}(G, I)$ when χ_i have conductor $\mathfrak{p}$ (depth zero): $\pi(\chi_1, \chi_2)$ has a nonzero I -fixed vector.

For deeper ramification (conductor $\mathfrak{p}^n$, $n \geq 2$): need even smaller compact open $K_n \subset I$.

Beyond Satake: Steinberg (Monodromy Enters)

St = unique irreducible quotient of $\text{Ind}_B^{\text{GL}_2}(|\cdot|^{1/2} \otimes |\cdot|^{-1/2})$.

$\text{St}^K = 0$, but $\text{St}' = \mathbb{C}$ (1-dimensional Iwahori-fixed space).

L-parameter:

$$\varphi_{\text{St}}(\text{Frob}) = \begin{pmatrix} p^{1/2} & 0 \\ 0 & p^{-1/2} \end{pmatrix}, \quad \varphi_{\text{St}}|_{\text{SL}_2} = \text{id}$$

$$\varphi_{\text{St}}\left(\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}\right) = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} \quad (\text{nilpotent monodromy})$$

The **same** $\varphi(\text{Frob}) = \text{diag}(p^{1/2}, p^{-1/2})$ with $\varphi|_{\text{SL}_2} = 1$ gives the *reducible* principal series. The SL_2 factor distinguishes the irreducible quotient.

Beyond Satake: Supercuspidal Representations

Cannot be obtained from parabolic induction from any proper Levi.

For $\mathrm{GL}_2(\mathbb{Q}_p)$: let $E/\mathbb{Q}_p$ be quadratic, $\theta : E^\times \rightarrow \mathbb{C}^\times$ not factoring through $\mathrm{Nm}_{E/F}$.

$$\pi_\theta = \mathrm{c}\text{-Ind}_{E^\times}^{\mathrm{GL}_2(\mathbb{Q}_p)} \theta \quad (\text{supercuspidal})$$

L-parameter: **irreducible** 2-dimensional representation

$$\varphi_\theta = \mathrm{Ind}_{W_E}^{W_{\mathbb{Q}_p}} \theta : W_{\mathbb{Q}_p} \rightarrow \mathrm{GL}_2(\mathbb{C})$$

$\varphi_\theta|_{\mathrm{SL}_2} = 1$ but $\varphi_\theta|_{I_{\mathbb{Q}_p}}$ nontrivial and φ_θ irreducible.

$\pi_\theta^K = 0$, $\pi_\theta^I = 0$ (for deep ramification) — invisible to both $\mathcal{H}(G, K)$ and $\mathcal{H}(G, I)$.

Summary Table for GL_2

π	$\varphi(\text{Frob})$	$\varphi _{I_F}$	$\varphi _{\text{SL}_2}$	Visible to
Unram. p.s.	$\text{diag}(\alpha, \beta)$	triv	triv	$\mathcal{H}(G, K)$
Depth-0 ram. p.s.	$\chi_1 \oplus \chi_2$	ntriv	triv	$\mathcal{H}(G, I)$
Steinberg	$\text{diag}(p^{1/2}, p^{-1/2})$	triv	id	$\mathcal{H}(G, I)$
Supercusp. (d.0)	$\text{Ind}_{W_E}^{W_F} \theta$	ntriv	triv	$\mathcal{H}(G, I)$
Supercusp. (deep)	$\text{Ind}_{W_E}^{W_F} \theta$	wild	triv	deeper level

Beyond GL_n : Nontrivial L-packets

For $G = \mathrm{SL}_2$, $\hat{G} = \mathrm{PGL}_2(\mathbb{C})$. Consider $\varphi : W_F \rightarrow \mathrm{PGL}_2(\mathbb{C})$ lifting to $\tilde{\varphi}(\mathrm{Frob}) = \mathrm{diag}(\alpha, -\alpha)$.

$$C_\varphi = \mathrm{Cent}_{\mathrm{PGL}_2}(\mathrm{Im} \varphi) \cong \mathbb{Z}/2\mathbb{Z}, \quad S_\varphi = \mathbb{Z}/2\mathbb{Z}$$

$\Rightarrow$ L-packet Π_φ contains **two** representations of $\mathrm{SL}_2(F)$.

Three axes of complexity in LLC:

1. **Ramification** ($\varphi|_{I_F}$): trivial $\rightarrow$ tame $\rightarrow$ wild
2. **Monodromy** ($\varphi|_{\mathrm{SL}_2}$): trivial $\rightarrow$ nontrivial
3. **L-packet structure** (S_φ): trivial (GL_n) $\rightarrow$ nontrivial (general G)

Satake sits at the **origin** of all three axes.

From Classical to Categorical: The Full Circle



For Satake:

- Classical: $\mathcal{H}_G \cong R(\hat{G})$
- Categorical: $\text{Perv}_{L+G}(\text{Gr}_G) \simeq \text{Rep}(G^\vee)$
- Decategorification: $\text{Rep}(\hat{G}) \xrightarrow{\text{Tr}_\sigma} \text{Coh}_{\text{fr}}^{\hat{G}}(\hat{G}\sigma) \xrightarrow{\text{End}(\tilde{\mathbf{I}})} R(\hat{G}) \cong \mathcal{H}_G$

For the full LLC, each axis corresponds to a geometric refinement:

- **Ramification** $\leftrightarrow$ relaxing equivariance ($L^+G \rightarrow I \rightarrow \text{smaller}$)
- **Monodromy** $\leftrightarrow$ SL_2 -factor in categorical trace formalism
- **L-packet structure** $\leftrightarrow$ categorical structure (IndCoh vs. QCoh)

Geometric Satake and Generalizations

Review: The Affine Grassmannian

G connected reductive over k , $F = k((t))$, $\mathcal{O} = k[[t]]$.

$$\mathrm{Gr}_G = LG/L^+G \quad (\text{ind-projective ind-scheme})$$

For GL_n : an R -point of $\mathrm{Gr}_{\mathrm{GL}_n}$ is an R -family of **lattices** $\Lambda \subset R((t))^n$ with $\Lambda \otimes_{R[[t]]} R((t)) = R((t))^n$.

Standard lattice: $\Lambda_0 = R[[t]]^n$.

GL_2 : Schubert Cells on Gr

T = diagonal torus, B = upper triangular. Dominant coweights:
 $\lambda = (a, b)$, $a \geq b$.

λ	$\dim Gr^\lambda$	Description
$(0, 0)$	0	$\{\Lambda_0\}$ (standard lattice)
$(1, 0)$	1	$\{\Lambda : \Lambda_0 \subset \Lambda \subset t^{-1}\Lambda_0, \dim_k(\Lambda/\Lambda_0) = 1\} \cong \mathbb{P}^1$
$(1, 1)$	0	$\{t^{-1}\Lambda_0\}$ (scalar shift)
$(2, 0)$	2	$\Lambda_0 \subset \Lambda \subset t^{-2}\Lambda_0, \dim_k(\Lambda/\Lambda_0) = 2$

Cartan decomposition: $LG = \bigsqcup_{\lambda \in X_*(T)^+} L^+G \cdot t^\lambda \cdot L^+G$.

Dimension formula (Lecture 4): $\dim Gr^\lambda = \langle 2\rho, \lambda \rangle$ where
 $\rho = \frac{1}{2} \sum_{\alpha > 0} \alpha$.

For GL_2 : $2\rho = (1, -1)$, so $\dim Gr^{(a,b)} = a - b$.

The Affine Flag Variety

Recall: t is a uniformizer of $F = k((t))$, so $\mathcal{O} = k[[t]]$, $L^+G = G(k[[t]])$, $LG = G(k((t)))$.

$\mathrm{Gr}_G = LG/L^+G$ uses **hyperspecial** L^+G , captures only unramified reps.

Replace by **Iwahori** $I = \mathrm{ev}_0^{-1}(B) \subset L^+G$ (preimage of Borel at $t = 0$).

Definition

The *affine flag variety* is $\mathrm{Fl}_G = LG/I$, with $\pi : \mathrm{Fl}_G \rightarrow \mathrm{Gr}_G$ (fiber $L^+G/I \cong G/B$).

Modular interpretation (GL_n): $\mathrm{Fl}_{\mathrm{GL}_n}$ parametrizes **lattice chains**:

$$\mathrm{Fl}_{\mathrm{GL}_n}(R) = \left\{ (\Lambda_0 \supset \Lambda_1 \supset \cdots \supset \Lambda_{n-1} \supset t\Lambda_0) \left| \begin{array}{l} \text{each } \Lambda_i \text{ a lattice} \\ \Lambda_i/\Lambda_{i+1} \text{ loc. free rk } 1 \end{array} \right. \right\}$$

$\pi : \mathrm{Fl}_{\mathrm{GL}_n} \rightarrow \mathrm{Gr}_{\mathrm{GL}_n}$ sends the chain to Λ_0 .

Iwahori subgroup:

$$I = \left\{ \begin{pmatrix} a(t) & b(t) \\ c(t) & d(t) \end{pmatrix} \in \mathrm{GL}_2(k[[t]]) \mid c(0) = 0 \right\}$$

A k -point of $\mathrm{Fl}_{\mathrm{GL}_2}$: a pair (Λ, ℓ) where Λ is a lattice and $\ell \subset \Lambda/t\Lambda \cong k^2$ is a line (= complete flag in 2d fiber).

$\pi : \mathrm{Fl}_{\mathrm{GL}_2} \rightarrow \mathrm{Gr}_{\mathrm{GL}_2}$ forgets ℓ . Fiber = $\mathbb{P}(\Lambda/t\Lambda) \cong \mathbb{P}^1$.

Gr_G : Schubert cells indexed by $X_*(T)^+ = \{(a, b) : a \geq b\}$

Fl_G : cells indexed by $\widetilde{W}_{\mathrm{aff}} = X_*(T) \rtimes W$ (**no folding**)

Over $\mathrm{Gr}^{(1,0)} \cong \mathbb{P}^1$: the flag variety refines into **two** I -orbits.

Why the Iwahori Step Is Fundamental

$\mathcal{H}(G, I) = C_c^\infty(I \backslash G(F) / I)$ occupies a privileged position:

1. **Generators and relations.** T_s for simple affine reflections $s \in W_{\text{aff}}$:

$$(T_s - q)(T_s + 1) = 0 \quad + \text{ braid relations}$$

This is the **Iwahori–Hecke algebra** of Kazhdan–Lusztig theory.

2. **Bernstein’s theorem.** $Z(\mathcal{H}(G, I)) \cong \mathbb{C}[X_*(T)]^W \cong \mathcal{H}(G, K)$
Satake $\hookrightarrow$ Iwahori–Hecke **as the center**.
3. **Coverage.** I -fixed vectors: unramified, depth-zero ramified p.s., Steinberg, depth-zero supercuspidals.
4. **Beyond Iwahori.** $\mathcal{H}(G, K_n)$ (Moy–Prasad) has no KL basis, no quadratic relation $\Rightarrow$ wild ramification largely open.

$\mathcal{H}(G, I)$ is **not commutative** (unlike $\mathcal{H}(G, K)$).

Review: IC Sheaves on Gr_G

The IC sheaf on the Schubert variety $\overline{\mathrm{Gr}}^\lambda$:

$$\mathrm{IC}_\lambda = j_{!*}(\underline{k}_{\mathrm{Gr}^\lambda}[\langle 2\rho, \lambda \rangle])$$

$j : \mathrm{Gr}^\lambda \hookrightarrow \overline{\mathrm{Gr}}^\lambda$ open, $j_{!*}$ = intermediate extension.

Characterized by: (1) $\mathrm{IC}_\lambda|_{\mathrm{Gr}^\lambda} = \underline{k}[\langle 2\rho, \lambda \rangle]$, (2) support condition, (3) cosupport condition.

Example (GL_2):

- $\mathrm{IC}_{(0,0)} = k_{\{L_0\}}$ (skyscraper at origin)
- $\mathrm{IC}_{(1,0)} = \underline{k}_{\mathbb{P}^1}[1]$ (constant sheaf on $\mathbb{P}^1$, shifted)
- $\mathrm{IC}_{(1,1)} = k_{\{t^{-1}\Lambda_0\}}$ (skyscraper)
- $\mathrm{IC}_{(2,0)}$: IC of $\overline{\mathrm{Gr}}^{(2,0)}$, *not* shifted constant (singular!)

Simple objects of $\mathrm{Perv}_{L+G}(\mathrm{Gr}_G) = \{\mathrm{IC}_\lambda\}_{\lambda \in X_*(T)^+}$.

Review: Convolution Product

(Lecture 6; [BR17, §4–5])

$$\mathcal{F}_1 \star \mathcal{F}_2 = m_! \left((q^*)^{-1} p^* (\mathcal{F}_1 \boxtimes \mathcal{F}_2) \right)$$

via the convolution diagram:

$$\mathrm{Gr}_G \times \mathrm{Gr}_G \xleftarrow{p} LG \times \mathrm{Gr}_G \xrightarrow{q} LG \times^{L^+G} \mathrm{Gr}_G \xrightarrow{m} \mathrm{Gr}_G$$

Example (GL_2):

$$\mathrm{IC}_{(1,0)} \star \mathrm{IC}_{(1,0)} = \mathrm{IC}_{(2,0)} \oplus \mathrm{IC}_{(1,1)}$$

corresponding to $\mathrm{std} \otimes \mathrm{std} = \mathrm{Sym}^2 \oplus \det$ in $\mathrm{Rep}(\mathrm{GL}_2^\vee)$. ✓

IC Sheaves and Convolution on Fl_G

Generalization: on Fl_G , simple objects of $\mathrm{Perv}_I(\mathrm{Fl}_G)$ are IC_w , indexed by $w \in W_{\mathrm{aff}}$ — **infinitely many** more than on Gr_G .

Convolution $\star$ replaces L^+G by I : twisted product $LG \times^I \mathrm{Fl}_G$.

Crucial failure: $\mathrm{Perv}_I(\mathrm{Fl}_G)$ is **not semisimple!**

$\exists$ nontrivial $\mathrm{Ext}^1(\mathrm{IC}_w, \mathrm{IC}_{w'}) \neq 0$.

Parity vanishing (Lecture 7) relied on Gr^λ being an affine bundle over a partial flag variety — this breaks for Iwahori orbits.

Rep-theoretic counterpart: Steinberg sits in a nonsplit exact sequence

$$0 \rightarrow \mathbf{1} \rightarrow \mathrm{Ind}_B^G(| \cdot |^{1/2} \otimes | \cdot |^{-1/2}) \rightarrow \mathrm{St} \rightarrow 0$$

Review: Fiber Functor and Tannakian Reconstruction

(Lectures 7–8; [BR17, Theorem 5.9])

The fiber functor $F = \bigoplus_{\mu} F_{\mu} : \text{Perv}_{L+G}(\text{Gr}_G) \rightarrow \text{Vect}_k$:

$$F_{\mu}(\mathcal{A}) = H_{T_{\mu}}^{\langle 2\rho, \mu \rangle}(\text{Gr}_G, \mathcal{A}) \cong H_c^{\langle 2\rho, \mu \rangle}(S_{\mu}, \mathcal{A})$$

$S_{\mu} = N((t)) \cdot t^{\mu}$ (semi-infinite orbit). Parity vanishing $\Rightarrow$ direct sum decomposition.

Example (GL_2): $\text{IC}_{(1,0)} \leftrightarrow \text{std}$: weights $(1, 0), (0, 1)$, each $F_{\mu} = k$.

$\text{IC}_{(2,0)} \leftrightarrow \text{Sym}^2$: weights $(2, 0), (1, 1), (0, 2)$, each multiplicity 1. $\checkmark$

Tannakian reconstruction (Lecture 8; [DM82]):

$$\text{Perv}_{L+G}(\text{Gr}_G, k) \simeq \text{Rep}_k(G_k^{\vee})$$

What Replaces $\text{Rep}(G^\vee)$ at Deeper Levels?

Tannakian formalism requires: (i) semisimple abelian category, (ii) fiber functor to Vect .

At Iwahori level: $\text{Perv}_I(\text{Fl}_G)$ fails (i) $\Rightarrow$ Tannakian does not apply.

Instead: work at the **derived/DG categorical** level.

Level	Automorphic	Spectral
L^+G (hyperspecial)	$\text{Perv}_{L^+G}(\text{Gr}_G) \simeq \text{Rep}(G^\vee)$	Tannakian
I (Iwahori)	$D_I(\text{Fl}_G)$	$\text{Coh}(\hat{\mathfrak{g}}^*/\hat{G})$
Deeper	$D_{K_n}(\text{Fl}_G)$	? (open)

At Iwahori: spectral side involves coherent sheaves on dual Lie algebra or Steinberg variety — accommodates extensions and monodromy [AB09].

Classical Motivation for the Trace

For an algebra A over E :

$$\mathrm{Tr}(A) = A/[A, A]$$

Universal trace function: any $f : A \rightarrow V$ with $f(ab) = f(ba)$ factors through Tr .

Equivalently: $\mathrm{Tr}(A) = \text{coequalizer of } A \otimes A \rightrightarrows A$ given by $a \otimes b \mapsto ab$ and $a \otimes b \mapsto ba$.

Categorical Trace: Definition

Let $(\mathcal{C}, \otimes, \mathbf{1})$ be an E -linear monoidal category, $\sigma : \mathcal{C} \rightarrow \mathcal{C}$ a monoidal endofunctor.

Definition (Categorical trace; [Zhu18, §4]; Lecture 9)

$\mathrm{Tr}_\sigma(\mathcal{C})$ is the 2-colimit:

$$\mathcal{C}^{\otimes 3} \begin{matrix} \rightarrow \\ \rightarrow \\ \rightarrow \end{matrix} \mathcal{C}^{\otimes 2} \rightrightarrows \mathcal{C}$$

Two functors $\mathcal{C}^{\otimes 2} \rightarrow \mathcal{C}$: $(X, Y) \mapsto X \otimes Y$ and $(X, Y) \mapsto Y \otimes \sigma X$.

Concretely: Objects of $\mathrm{Tr}_\sigma(\mathcal{C}) =$ objects of $\mathcal{C}$ (denoted $\tilde{X}$). Morphisms:

$$\mathrm{Hom}_{\mathrm{Tr}_\sigma}(\tilde{X}, \tilde{Y}) = \left(\bigoplus_{(V, W)} \mathrm{Hom}(X, V \otimes W) \otimes_E \mathrm{Hom}(W \otimes \sigma V, Y) \right) / \sim$$

Key: in $\mathrm{Tr}_\sigma(\mathcal{C})$, $\widetilde{X \otimes Y} \cong \widetilde{Y \otimes \sigma X}$ — enforces “ σ -twisted commutativity.”

The Key Computation: $\mathrm{Tr}_\sigma(\mathrm{Rep}(\hat{G}))$

$\hat{G}$ Langlands dual, $\sigma : \hat{G} \rightarrow \hat{G}$ q -Frobenius (for split G : $\sigma = \mathrm{id}$).

σ -twisted conjugation: $c_\sigma(g) \cdot h = g^{-1}h\sigma(g)$, $g \in \hat{G}$, $h \in \hat{G}\sigma$.

$\mathrm{Coh}^{\hat{G}}(\hat{G}\sigma) = \hat{G}$ -equivariant coherent sheaves on $[\hat{G}\sigma/\hat{G}]$.

For $V \in \mathrm{Rep}(\hat{G})$: define $\tilde{V} \in \mathrm{Coh}^{\hat{G}}(\hat{G}\sigma)$ with global sections $\mathcal{O}_{\hat{G}} \otimes V$, $\hat{G}$ -action $g \cdot f(h) = g(f(g^{-1}h\sigma(g)))$.

$\mathrm{Coh}_{\mathrm{fr}}^{\hat{G}}(\hat{G}\sigma) =$ full subcategory spanned by such $\tilde{V}$.

Proposition ([Lee26, Prop. 2.4]; [Zhu18])

$\mathrm{Tr}_\sigma(\mathrm{Rep}(\hat{G})) \cong \mathrm{Coh}_{\mathrm{fr}}^{\hat{G}}(\hat{G}\sigma)$, sending $\tilde{V} \mapsto \tilde{V}$.

$\mathrm{Hom}_{\mathrm{Coh}_{\mathrm{fr}}^{\hat{G}}}(\tilde{V}_1, \tilde{V}_2) = (\mathcal{O}_{\hat{G}} \otimes V_1^* \otimes V_2)^{\hat{G}}$.

Local Shtukas and the Decategorification Theorem

Define the moduli of **local Shtukas**:

$$\mathrm{Sht}^{\mathrm{loc}} = LG / \mathrm{Ad}_\sigma L^+ G$$

(LG modulo σ -twisted conjugation $g \mapsto h^{-1}g\sigma(h)$ by $h \in L^+ G$).

Notation guide:

- $\mathrm{Hk}^{\mathrm{loc}} = \mathrm{Gr}_G \times_{\mathrm{Gr}_G} \mathrm{Gr}_G$: local Hecke stack (parametrizes pairs of modifications)
- $\mathrm{P}(\mathrm{Hk}^{\mathrm{loc}})$: perverse sheaves on $\mathrm{Hk}^{\mathrm{loc}}$; equiv. to $\mathrm{Perv}_{L^+ G}(\mathrm{Gr}_G)$ via Satake
- Φ^{loc} : the *local Frobenius* map $\mathrm{Hk}^{\mathrm{loc}} \rightarrow \mathrm{Sht}^{\mathrm{loc}}$ (sends a modification to its Frobenius twist)
- $\mathrm{P}^{\mathrm{Hk}}(\mathrm{Sht}^{\mathrm{loc}})$: perverse sheaves on $\mathrm{Sht}^{\mathrm{loc}}$ equipped with Hecke structure (= action of $\mathrm{Rep}(\hat{G})$ via cohomological correspondences)

$\delta_1 \in \mathrm{P}^{\mathrm{Hk}}(\mathrm{Sht}^{\mathrm{loc}})$: image of $\mathrm{IC}_0 \in \mathrm{Perv}_{L^+ G}(\mathrm{Gr})$ under

$$\mathrm{Perv}_{L^+ G}(\mathrm{Gr}) \xrightarrow{\sim} \mathrm{P}(\mathrm{Hk}^{\mathrm{loc}}) \xrightarrow{\Phi^{\mathrm{loc}}} \mathrm{P}(\mathrm{Sht}^{\mathrm{loc}}).$$

Theorem ([Zhu18, Thm 4.3.1]; [Lee26, Thm 4.9])

(1) $\exists$ fully faithful $S : \mathrm{Coh}_{\mathrm{fr}}^{\hat{G}}(\hat{G}\sigma) \rightarrow \mathrm{P}^{\mathrm{Hk}}(\mathrm{Sht}^{\mathrm{loc}})$.

(2) $S(\mathcal{O}_{[\hat{G}\sigma/\hat{G}]}) = \delta_1$, and on endomorphisms:

$$S : \overline{\mathbb{Q}}_\ell[\hat{G}\sigma]^{\hat{G}} = \mathrm{End}(\tilde{\mathbf{1}}) \longrightarrow \mathrm{End}(\delta_1) \cong \mathcal{H}_G \otimes \overline{\mathbb{Q}}_\ell$$

is the **classical Satake isomorphism**.

The Commutative Diagram

$$\begin{array}{ccc}
 \mathrm{Rep}(\hat{G}) & \xrightarrow[\sim]{\mathrm{Sat}} & \mathrm{P}(\mathrm{Hk}^{\mathrm{loc}}) \\
 \downarrow V \mapsto \tilde{V} & & \downarrow \Phi^{\mathrm{loc}} \\
 \mathrm{Coh}_{\mathrm{fr}}^{\hat{G}}(\hat{G}\sigma) & \xrightarrow{S} & \mathrm{P}^{\mathrm{Hk}}(\mathrm{Sht}^{\mathrm{loc}})
 \end{array}$$

Top row: categorical Geometric Satake (equivalence of categories)

Left column: categorical trace ($V \mapsto \tilde{V}$)

Decategorification: take $\mathrm{End}(\tilde{\mathbf{1}})$:

$$\underbrace{\mathrm{Rep}(\hat{G})}_{\text{category}} \xrightarrow{\mathrm{Tr}_{\sigma}} \underbrace{\mathrm{Coh}_{\mathrm{fr}}^{\hat{G}}(\hat{G}\sigma)}_{\text{category}} \xrightarrow{\mathrm{End}(\tilde{\mathbf{1}})} \underbrace{R(\hat{G}) \cong \mathcal{H}_G}_{\text{algebra}}$$

For split G : $\mathrm{End}(\tilde{\mathbf{1}}) = \overline{\mathbb{Q}}_{\ell}[\hat{G}]^{\hat{G}} = R(\hat{G}) \otimes \overline{\mathbb{Q}}_{\ell}$.

Decategorification at Each Level

Theorem (Zhu [Zhu25, Thm 1.6])

The categorical trace extends to Iwahori/tame level. The σ -twisted trace of the Iwahori-level equivalence recovers the tame LLC.

Categorical (geom. side)	$\xrightarrow{\text{Tr}_\sigma}$	Classical LLC
$\text{Perv}_{L+G}(\text{Gr}_G) \simeq \text{Rep}(G^\vee)$	$\rightarrow$	$\mathcal{H}(G, K) \cong R(\hat{G})$
$D_I(\text{Fl}_G) \simeq (\text{spectral})$	$\rightarrow$	$\mathcal{H}(G, I) \leftrightarrow \text{tame LLC}$
$D_{K_n}(?) \simeq ?$	$\rightarrow$	wild LLC (open)

Open Problem. Extend categorical LLC to wild ramification. Difficulty: wild inertia destroys semisimplicity at all levels.

Global Geometric Langlands

Derived Categories and D-modules: Setup

In Lectures 1–9 we worked with:

- $D_c^b(Y, \overline{\mathbb{Q}}_\ell)$ = bounded derived category of **constructible ℓ -adic sheaves** on Y
- $\text{Perv}(Y) \subset D_c^b(Y)$: the heart of the perverse t-structure (Lecture 5)

For the global story, we write $D(Y)$ to mean one of two settings:

Setting	$D(Y)$	Objects
$X/\mathbb{F}_q$ (ℓ -adic)	$D_c^b(Y, \overline{\mathbb{Q}}_\ell)$	constructible ℓ -adic sheaves
$X/\mathbb{C}$ (de Rham)	$\text{D-mod}(Y)$	algebraic D-modules on Y

A **D-module** on Y is a sheaf of modules over the sheaf $\mathcal{D}_Y$ of differential operators — equivalently, a quasi-coherent sheaf with a flat connection ∇ . Example: a local system (V, ∇) on Y is a D-module.

Riemann–Hilbert ($Y/\mathbb{C}$, smooth): $D_{\text{hol}}^b(\text{D-mod}(Y)) \xrightarrow{\sim} D_c^b(Y, \mathbb{C})$
sends $(V, \nabla) \mapsto \ker \nabla$ (local system of flat sections). Under RH, perverse sheaves $\leftrightarrow$ holonomic D-modules.

The Local–Global Dictionary

Local (at $x \in X$)	Global (on curve X)
Local field $F_x = k((t))$	Function field $F_X = k(X)$
$\mathrm{Gr}_G = LG/L^+G$	$\mathrm{Bun}_G(X)$
L-parameter $\varphi_x : W_{F_x} \rightarrow {}^L G$	$\sigma : \pi_1(X) \rightarrow G^\vee$
$\mathrm{Perv}_{L^+G}(\mathrm{Gr}_G) \simeq \mathrm{Rep}(G^\vee)$	$D(\mathrm{Bun}_G) \simeq \mathrm{IndCoh}(\mathrm{LocSys}_{G^\vee})$

Local-global compatibility: σ on X restricts at each x to $\sigma|_{W_{F_x}}$.

If σ unramified at x : $\sigma|_{W_{F_x}}$ is unramified $\Rightarrow$ Satake governs.

At finitely many “bad” points: need deeper level structures from §3.

X smooth projective curve over $\mathbb{F}_q$ (or $\mathbb{C}$).

$$\text{Bun}_G(X) = \{\text{principal } G\text{-bundles on } X\} \quad (\text{algebraic stack})$$

- Smooth, dimension $(g-1)\dim G$
- Bun_{GL_n} = moduli of rank- n vector bundles
- $\pi_0(\text{Bun}_G) \cong \pi_1(G)$

Bun_G is the **geometric analog** of $G(F) \backslash G(\mathbb{A}_F) / \prod_v K_v$.

Example (GL_2 on $\mathbb{P}^1$): by Grothendieck, $\mathcal{E} \cong \mathcal{O}(a) \oplus \mathcal{O}(b)$, $a \geq b$.

$$\text{LocSys}_{G^\vee}(X) = \{\pi_1^{\text{ét}}(X) \rightarrow G^\vee(\overline{\mathbb{Q}_\ell})\} / G^\vee\text{-conj}$$

Why $D(\mathrm{Bun}_G) = \text{Categorified Automorphic Forms}$

Weil uniformization (over $\mathbb{F}_q$):

$$\mathrm{Bun}_G(\mathbb{F}_q) \cong G(F_X) \backslash G(\mathbb{A}_{F_X}) / \prod_v G(\mathcal{O}_v)$$

Functions on the right-hand side = **(spherical) automorphic forms**.

Function-sheaf dictionary (Grothendieck): a sheaf $\mathcal{F} \in D_c^b(\mathrm{Bun}_G, \overline{\mathbb{Q}}_\ell)$ yields a function on $\mathrm{Bun}_G(\mathbb{F}_q)$:

$$f_{\mathcal{F}}(x) = \sum_i (-1)^i \mathrm{Tr}(\mathrm{Frob}_x, H_x^i(\mathcal{F}))$$

Hecke eigensheaf $\Rightarrow$ Hecke eigenform: if $T_V(\mathcal{F}) \simeq V_\sigma \boxtimes \mathcal{F}$, then

$$T_{V,x}(f_{\mathcal{F}}) = \mathrm{Tr}(\mathrm{Frob}_x, (V_\sigma)_x) \cdot f_{\mathcal{F}}$$

(the Hecke eigenvalue at x is $\mathrm{Tr}(\mathrm{Frob}_x, (V_\sigma)_x)$, recovering classical Langlands).

$D(\mathrm{Bun}_G)$ is thus a **categorification** of the space of automorphic forms: *sheaves* are categorified *functions*, and the Hecke eigensheaf property categorifies the Hecke eigenvalue condition.

Definition (Hecke stack at x)

$$\mathrm{Hk}_x = \left\{ (\mathcal{E}, \mathcal{E}', \alpha) \mid \mathcal{E}, \mathcal{E}' \in \mathrm{Bun}_G, \alpha : \mathcal{E}|_{X \setminus \{x\}} \xrightarrow{\sim} \mathcal{E}'|_{X \setminus \{x\}} \right\}$$

By Beauville–Laszlo [BL95]: fibers of $\mathrm{Hk}_x \rightarrow \mathrm{Bun}_G$ are $\cong \mathrm{Gr}_G$.

Example (GL_2 on $\mathbb{P}^1$): $\mathcal{E} = \mathcal{O} \oplus \mathcal{O}$. A Hecke modification of type $(1, 0)$ at $x = 0$: choose a line $\ell \subset \mathcal{E}_0 \cong k^2$, giving $\mathcal{E}' = \mathcal{O}(1) \oplus \mathcal{O}(-1)$. The modification is parametrized by $\mathbb{P}^1$ (the choice of ℓ).

Hecke Operators from Geometric Satake

For $V \in \text{Rep}(G^\vee)$, Geometric Satake gives $\text{IC}_V \in \text{Perv}_{L+G}(\text{Gr}_G)$.

Definition (Hecke functor)

The Hecke stack $\text{Hk}_G = \{(\mathcal{E}, \mathcal{E}', x, \alpha)\}$ has three projections:

$$\text{Bun}_G \xleftarrow{\text{pr}_1} \text{Hk}_G \xrightarrow{\text{pr}_2 \times \text{supp}} \text{Bun}_G \times X$$

$$\text{pr}_1(\mathcal{E}, \mathcal{E}', x, \alpha) = \mathcal{E}, \text{pr}_2(\mathcal{E}, \mathcal{E}', x, \alpha) = \mathcal{E}', \text{supp}(\mathcal{E}, \mathcal{E}', x, \alpha) = x.$$

The **global** Hecke operator (varying the modification point over X):

$$T_V : D(\text{Bun}_G) \rightarrow D(X \times \text{Bun}_G), \quad T_V(\mathcal{F}) = (\text{pr}_2 \times \text{supp})_! (\text{pr}_1^*(\mathcal{F}) \otimes \text{IC}_V)$$

Composition: at a fixed $x \in X$, the fiber $T_{V,x} : D(\text{Bun}_G) \rightarrow D(\text{Bun}_G)$ satisfies $T_{V,x} \circ T_{W,x} \simeq T_{V \otimes W,x}$ (from convolution on Gr_G).

Globally: T_W lands in $D(X \times \text{Bun}_G)$; applying T_V along Bun_G gives $D(X \times X \times \text{Bun}_G)$. Restricting to the diagonal $\Delta(X) \subset X \times X$ recovers $T_{V \otimes W}$.

Hecke Eigensheaves

Given $\sigma : \pi_1(X) \rightarrow G^\vee$ (a $G^\vee$ -local system on X) and $V \in \text{Rep}(G^\vee)$, define V_σ : the **local system on X** obtained by composing σ with $G^\vee \rightarrow \text{GL}(V)$. Concretely, V_σ is the vector bundle with flat connection on X whose monodromy is σ acting through V .

Definition

$\mathcal{F} \in D(\text{Bun}_G)$ is a **Hecke eigensheaf** with eigenvalue $\sigma \in \text{LocSys}_{G^\vee}(X)$ if for all $V \in \text{Rep}(G^\vee)$:

$$T_V(\mathcal{F}) \simeq V_\sigma \boxtimes \mathcal{F} \quad \text{in } D(X \times \text{Bun}_G)$$

Here $\boxtimes$ is the **external tensor product**:

$V_\sigma \boxtimes \mathcal{F} := \text{pr}_X^*(V_\sigma) \otimes \text{pr}_{\text{Bun}}^*(\mathcal{F})$, a sheaf on $X \times \text{Bun}_G$. At each $x \in X$, this reads $T_{V,x}(\mathcal{F}) \simeq (V_\sigma)_x \otimes \mathcal{F}$.

Analogy: $T_p f = \lambda_p f$ for classical Hecke eigenforms. The “eigenvalue” is not a number but a **local system on X** .

Function-sheaf dictionary: $\mathcal{F} \mapsto f_{\mathcal{F}}$ with

Localization I: Automorphic Side (Hecke at x)

Fix $x \in X$. How does $\text{Rep}(\hat{G})$ act on $D(\text{Bun}_G)$?

Step 1: The fiber of the Hecke stack Hk_G over x is a twist of $\text{Gr}_{G,x}$:

$$(h^\rightarrow)^{-1}(x, \mathcal{E}') \cong \mathcal{E}'(\mathcal{O}_x) \times^{G(\mathcal{O}_x)} \text{Gr}_{G,x}$$

Step 2: Geometric Satake at x :

$$\text{Sat}(G)_x : \text{Rep}(\hat{G}) \xrightarrow{\sim} \text{Perv}_{L+G}(\text{Gr}_{G,x})$$

sends $V \in \text{Rep}(\hat{G})$ to $\text{IC}_V \in \text{Perv}_{L+G}(\text{Gr}_{G,x})$.

Step 3: The Hecke functor at x (domain = codomain = $D(\text{Bun}_G)$):

$$T_{V,x}(\mathcal{F}) = (\text{pr}_2)_! (\text{pr}_1^*(\mathcal{F}) \otimes \text{IC}_V) \in D(\text{Bun}_G)$$

This defines a **monoidal action**: $\text{Rep}(\hat{G}) \rightarrow \text{End}(D(\text{Bun}_G))$, $V \mapsto T_{V,x}$,

with $T_{V,x} \circ T_{W,x} \simeq T_{V \otimes W,x}$ (from the convolution on $\text{Gr}_{G,x}$).

Localization II: Spectral Side (Evaluation at x)

Fix $x \in X$. How does $\text{Rep}(\hat{G})$ act on $\text{IndCoh}(\text{LocSys}_{\hat{G}})$?

Step 1: A point of $\text{LocSys}_{\hat{G}}$ is a local system $\sigma : \pi_1(X) \rightarrow \hat{G}$.

Step 2: Fix $V \in \text{Rep}(\hat{G})$. Evaluation at x defines a quasi-coherent sheaf $\text{ev}_x(V)$ on $\text{LocSys}_{\hat{G}}$:

$$\text{ev}_x(V)|_{\sigma} = (V_{\sigma})_x = \text{fiber of the associated local system } V_{\sigma} \text{ at } x$$

This is a vector bundle on $\text{LocSys}_{\hat{G}}$ whose fiber over σ is the vector space V with $\hat{G}$ -action twisted by σ at x .

Step 3: $\text{ev}_x(V) \in \text{QCoh}(\text{LocSys}_{\hat{G}})$ acts on $\text{IndCoh}(\text{LocSys}_{\hat{G}})$ by:

$$\mathcal{G} \longmapsto \text{ev}_x(V) \otimes \mathcal{G}$$

This is again monoidal: $\text{ev}_x(V) \otimes \text{ev}_x(W) \cong \text{ev}_x(V \otimes W)$.

Example: σ irreducible, $\mathcal{G} = \delta_{\sigma}$ (skyscraper at σ). Then $\text{ev}_x(V) \otimes \delta_{\sigma} \cong (V_{\sigma})_x \otimes \delta_{\sigma}$, a skyscraper tensored with a vector space.

Localization III: Property He and Geometric Satake

Both sides of GLC carry an action of $\text{Rep}(\hat{G})$ at each $x \in X$:

Automorphic: $V \mapsto T_{V,x}$ Spectral: $V \mapsto \text{ev}_x(V) \otimes (-)$

Property He (Hecke compatibility; [GR+24, Ga15])

The Langlands functor $L_G : \text{D-mod}(\text{Bun}_G) \xrightarrow{\sim} \text{IndCoh}(\text{LocSys}_{\hat{G}})$
intertwines these two actions: for every $V \in \text{Rep}(\hat{G})$ and $x \in X$,

$$L_G(T_{V,x}(\mathcal{F})) \simeq \text{ev}_x(V) \otimes L_G(\mathcal{F})$$

What this says: the Hecke operator $T_{V,x}$ on the automorphic side (defined via IC_V on $\text{Gr}_{G,x}$, i.e., Geometric Satake) corresponds to tensoring by $\text{ev}_x(V)$ on the spectral side. So:

$$\boxed{\text{Geometric Satake at } x} = \boxed{\text{local fiber of GLC at } x}$$

The global equivalence is assembled from copies of Geometric Satake at all $x \in X$, organized by the **Ran space**

The Spectral Side: QCoh vs. IndCoh

Recall: $\mathrm{D}\text{-mod}(\mathrm{Bun}_G)$ is the **automorphic side** (defined in the Setup slide). The **spectral side** involves sheaves on $\mathrm{LocSys}_{\hat{G}}$:

$\mathrm{QCoh}(Y)$ = derived category of quasi-coherent sheaves on a stack Y .

$\mathrm{IndCoh}(Y)$ = ind-completion of $\mathrm{Coh}(Y)$ (bounded coherent sheaves).

Key distinction:

- Y **smooth**: $\mathrm{IndCoh}(Y) \simeq \mathrm{QCoh}(Y)$ (no difference)
- Y **singular**: $\mathrm{IndCoh}(Y) \neq \mathrm{QCoh}(Y)$. IndCoh detects the singularity; QCoh loses it.

$\mathrm{LocSys}_{\hat{G}}$ is typically **singular** (e.g., at reducible local systems, the stack has non-trivial derived structure), so IndCoh is essential.

Analogy: IndCoh is to QCoh as distributions are to smooth functions — IndCoh includes “distributional” objects (skyscrapers at singular points) that QCoh cannot see.

The Geometric Langlands Conjecture

Conjecture (Beilinson–Drinfeld, Arinkin–Gaitsgory [AG15])

$$\mathrm{D}\text{-mod}(\mathrm{Bun}_G) \simeq \mathrm{IndCoh}(\mathrm{LocSys}_{\hat{G}})$$

(equivalence of DG categories)

- **Automorphic side:** $\mathrm{D}\text{-mod}(\mathrm{Bun}_G)$ — D-modules on the moduli of G -bundles
- **Spectral side:** $\mathrm{IndCoh}(\mathrm{LocSys}_{\hat{G}})$ — ind-coherent sheaves on the stack of $\hat{G}$ -local systems
- Hecke eigensheaves $\leftrightarrow$ skyscraper sheaves on $\mathrm{LocSys}_{\hat{G}}$

Theorem (Gaitsgory–Raskin–et al. [GR+24], 2024)

This holds in the de Rham setting $(X/\mathbb{C}, \text{any reductive } G)$.

Global with Level Structure

Without level: Hecke operators use IC_V on Gr_G
(hyperspecial/unramified).

To incorporate ramification: introduce **level structures** at finitely many points.

Local input at x	Global moduli	Hecke at x
$Gr_G = LG/L^+G$	Bun_G	T_V from $IC_V \in Perv_{L^+G}(Gr)$
$Fl_G = LG/I$	Bun_G^I	Hecke from $Perv_I(Fl_G)$

Iwahori level at x : replace Bun_G by G -bundles with a **Borel reduction** of the fiber $\mathcal{E}_x$, and use IC sheaves on Fl_G at x .

How Geometric Satake Enters the Global Picture

Geometric Satake is the **local input** for the global correspondence:

1. **Defines the dual group** $\hat{G}$ (Tannakian reconstruction from $\text{Perv}_{L+G}(\text{Gr}_G)$)
2. **Provides Hecke operators** (IC_V defines T_V)
3. **Starting point** of the level hierarchy (hyperspecial $\rightarrow$ Iwahori $\rightarrow \dots$)

Local-to-global embeddings ([Ga15]):

- Automorphic: $\text{D-mod}(\text{Bun}_G) \hookrightarrow \text{D-mod}(\text{Gr}_{G, \text{Ran}(X)})$ (fully faithful)
- Spectral: $\text{QCoh}(\text{LocSys}_{\hat{G}}) \hookrightarrow \text{Rep}(\hat{G})_{\text{Ran}(X)}$ (fully faithful, Gaitsgory–Lurie)

Both global categories embed into **Ran-space versions of Geometric Satake**. The GLC equivalence is compatible with these embeddings (Property He).

The p -adic Analog and Big Picture

Fargues–Scholze: p -adic Geometric Langlands

Replace the function field curve $X/\mathbb{F}_q$ by the **Fargues–Fontaine curve** X_{FF} : a “curve” over $\mathbb{Q}_p$ whose geometry encodes p -adic Hodge theory.

Theorem (Fargues–Scholze [FS21, Thm I.4.1])

$\text{Bun}_G(X_{\text{FF}})$ has strata indexed by $B(G)$ (Kottwitz set = isocrystal classes), with each stratum Bun_G^b satisfying $D(\text{Bun}_G^b) \simeq D(G_b(E))$, where G_b is the inner form of G associated to $b \in B(G)$.

Conjecture (Fargues–Scholze, Zhu, Hansen–Mann)

$$D(\text{Bun}_G(X_{\text{FF}})) \simeq \text{IndCoh}(\text{Par}_{G^\vee})$$

where $\text{Par}_{G^\vee}$ = stack of L -parameters $W_E \rightarrow {}^L G$ (the “parameter space”).

KIAS Categorical Langlands Summit (June 14–26, 2026, Seoul):
Hansen–Mann (GL_n), Zhu (tame case).

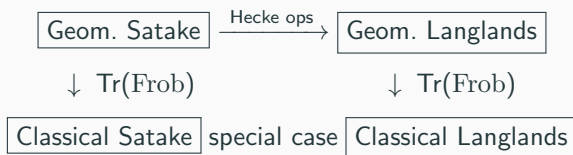
The Full Landscape

	Automorphic	Spectral
Classical local	$\mathcal{H}(G, K)$	$\mathbb{C}[\hat{T}/W]$
Geometric local	$\mathrm{Perv}_{L+G}(\mathrm{Gr}_G)$	$\mathrm{Rep}(\hat{G})$
Categorical local	$\mathrm{P}^{\mathrm{Hk}}(\mathrm{Sht}^{\mathrm{loc}})$	$\mathrm{Coh}_{\mathrm{fr}}^{\hat{G}}(\hat{G}\sigma)$
Geometric global	$\mathrm{D}\text{-mod}(\mathrm{Bun}_G)$	$\mathrm{IndCoh}(\mathrm{LocSys}_{\hat{G}})$

Each row is a **categorification** of the one above.

Each column is connected by duality.

Three Roles of Geometric Satake



1. **Special case** of LLC (hyperspecial level = unramified)
2. **Source of Hecke operators** (IC_V defines T_V)
3. **Starting point** of the level hierarchy (hyperspecial $\rightarrow$ Iwahori $\rightarrow \dots$)

- **Wild ramification:** No geometric description of LLC for arbitrary depth. Wild inertia destroys semisimplicity at all levels.
Frenkel–Gross rigid connection, Kaletha’s regular supercuspidals.
- **p -adic coefficients:** Replace $\overline{\mathbb{Q}}_\ell$ by $\overline{\mathbb{Z}}_\ell$ or $\overline{\mathbb{F}}_\ell$. Geometric Satake holds [MV07, Zhu], but semisimplicity fails $\Rightarrow$ modular representation theory enters.
- **Fargues–Scholze:** Geometrization of LLC via the Fargues–Fontaine curve — “ p -adic Geometric Langlands.”

Thank you!

Questions?